

Diego Renner

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 [DiegoRenner](#)

</> C++, Python, Rust

Date of Birth: 29.08.1995

Nationality: Swiss

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(Items relating to projects and papers are clickable.)



EDUCATION

- **Imperial College London** London, England
Aeronautics PhD May 2025 - now
 - Development of GPU Compressible Flow Solver for Turbomachinery (C++).
- **ETH Zurich** Zurich, Switzerland
M.Sc. Mathematics September 2021 - December 2023
 - Degree completed with a thesis on differentiable haemodynamics solver in JAX (Python).
- **ETH Zurich** Zurich, Switzerland
M.Sc. Computational Science and Engineering, Specialization Physics September 2018 - August 2021
 - Degree completed with a thesis on solving the transmission scattering problem using BEM (C++).
- **Universität Basel** Basel, Switzerland
B.Sc. Computational Mathematics September 2014 - December 2017
 - Completed extracurricular courses on Computer Architecture, Operating Systems and Quantum Mechanics.
- **Gymnasium Bäumlhof** Basel, Switzerland
Matura, Specialization Biology & Chemistry August 2009 - July 2014

PROJECTS & THESIS

- Parallelizing the Barnes-Hut Algorithm with MPI: Parallelized implementation of N-Body solver in C++ using the MPI framework. (Course Work)
- AiiDA Lab implementation of IR spectrum calculations for carbon based nanomaterials: An AiiDa workflow implemented in the Jupyter Notebooks based AiiDa lab interface. (Semesters Thesis, Computational Science)
- Near Resonances for Scattering Transmission Problems: A BEM based C++ solver for Scattering Transmission Problems, developed to investigate scatterer-dependent near resonances. (Master's Thesis, Computational Science)

- ML Based Game Simulation in a Finance Setting: Agents trained to trade or hold a stock taking into account real historical data on cash returns. Policies are learned via reinforcement learning. (Course Work)
- On differentiable simulations of haemodynamic systems: A 1D-haemodynamics solver written in Python using JAX. The differentiability of the solver aims to aid in the development of personalised medicine. (Master's Thesis, Mathematics)

PUBLICATIONS

- Detecting Near Resonances in Acoustic Scattering: Continued development of root finding algorithm from Master's Thesis using handcrafted optimization algorithm and state of the art computation of singular values. (Published)
- Accelerated Patient-Specific Calibration via Differentiable Haemodynamics Simulations: Demonstrating feasibility of parameter inference through differentiable 1D-haemodynamics solver written in Master's Thesis. (Preprint)
- Raman spectroscopy enabled automatic media release controlled by convolutional neural networks. (Work in Progress)

EXPERIENCE

- **Novartis Pharma AG** Basel, Switzerland
MSc Student Proc Dev & New Technologies *June 2024 - December 2024*
 - Developing ML/AI algorithms for classifying Raman spectroscopy data.**Technologies:** imbalanced-learn, JAX, Matplotlib, NumPy, scikit-learn, SciPy, TensorFlow
Theory: (C)NN, DoE, GMM, PCA, SMOTE
- **plantime.io** Basel, Switzerland
Software Engineer *January 2024 - April 2025*
 - Developing ML/AI algorithms for optimizing shift scheduling.**Technologies:** Rust
Theory: Evolutionary optimization algorithms
- **ETH Zurich** Zurich, Switzerland
Teaching Assistant *September 2021 - February 2022*
 - Teaching Assistant for Lecture "Numerical Methods for Computer Science".**Technologies:** C++
Theory: ODEs, PDEs and numerical algorithms to solve them
- **ETH Zurich** Zurich, Switzerland
Research Assistant *September 2020 - June 2021*
 - Hired for continued development of BEM code that was implemented in Master's Thesis.**Technologies:** C++, CMake, Git
Theory: BEM, Resonances in Transmission Scattering Problems
- **ETH Zurich** Zurich, Switzerland
Teaching Assistant *September 2020 - February 2021*
 - Teaching Assistant for Lecture "Numerical Methods".**Technologies:** C++, CMake
Theory: ODEs, PDEs and numerical algorithms to solve them
- **CSCS Swiss National Supercomputing Center** Lugano, Switzerland
Internship *May 2018 - August 2018*
 - Writing regression checks for Piz Daint, Cray XC40/XC50 production system.**Technologies:** C, MPI, MySQL, Kibana, Grafana

CERTIFICATES & EXTRACURRICULARS

- **Ready, set, go! A short introduction for Student Teaching Assistants** (remote) Zurich
Education Development and Technology, ETH Zurich April 2020
 - Improving didactic skills
 - Setting goals for upcoming teaching activity

- **Effective High-Performance Computing & Data Analytics with GPU** (remote) Lugano, Switzerland
Summerschool, CSCS-USI July 2020
 - GPU: architecture & programming (CUDA, OpenACC)
 - JupyterLab
 - Python: Numpy, SciPy, Dask, Numba
 - ML: Rapids
 - Deep Learning: TensorFlow

- **International Consulting Network (ICON)** Shanghai, (remote) Belo Horizonte
Student Consulting Network March 2017 - February 2018
 - Market Research & Trend Analysis consulting for CREP (Real Estate, China) & Lalubema (Private Security, Brazil)

VOLUNTEERING

- **Mapathon@Novartis with the Swiss Red Cross** Basel
Tracing maps in underserved areas for risk reduction and disaster response. October-November 2024

- **Jugendsession** Bern
Formulating policy suggestions to be passed on to the Parliament: Neutrality in Media. November 2011

- **Jugendsession** Bern
Formulating policy suggestions to be passed on to the Parliament: Electronic Voting. November 2009

NAMED REFERENCES

- **Dr. Andreas Jocksch**
Senior Research Software Engineer
 - Phone: +41 91 610 82 32
 - Mail: andreas.jocksch@cscs.ch

Relation: Supervisor during internship at CSCS on writing regression checks for Piz Daint, Cray XC40/XC50 production system.

- **Prof. Dr. Ralf Hiptmair**
Full Professor and Deputy head of Dep. of Mathematics / Head of Seminar for Applied Mathematics at ETH Zurich
 - Phone: +41 44 632 34 04
 - Mail: ralf.hiptmair@sam.math.ethz.ch

Relation: Supervisor of Computational Science Master's Thesis on solving the transmission scattering problem using BEM (C++).

- **Prof. Dr. Siddhartha Mishra**

Full Professor at the Dep. of Mathematics / Deputy head of Seminar for Applied Mathematics at ETH Zurich

- Phone: +41 44 632 75 63
- Mail: siddhartha.mishra@sam.math.ethz.ch

Relation: Supervisor of Mathematics Master's Thesis on differentiable haemodynamics solver in JAX (Python).

- **Dr. Georgios Kissas**

Postdoctoral Fellow at ETH AI Center + BAUG + MAVT + SAM

- Phone: +41 78 969 95 77
- Mail: gkissas@ai.ethz.ch

Relation: Co-Supervisor of Mathematics Master's Thesis on differentiable haemodynamics solver in JAX (Python) and Supervisor of publication thereof.